Multi-Resolution Semantic Abstraction over an Evolving Knowledge Hypergraph

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Abstract

Scientific knowledge continuously changes as new papers introduce new methods, datasets, techniques, and relationships.

Large scientific knowledge structures become difficult to analyse because they contain many heterogeneous entities and complex relationships.

This work studies a Temporal Knowledge Hypergraph (TKH) and proposes a multi-resolution semantic abstraction framework.

The proposed approach transforms the evolving hypergraph into an interpretable hierarchy by combining semantic information, higher-order relationships, and temporal evolution.

The framework contains temporal snapshot construction, semantic representation, a hypergraph-based abstraction objective, multi-resolution hierarchy construction, temporal coupling, and semantic labelling.

The usefulness of the generated hierarchy is evaluated through a downstream retrieval task.

Two retrieval settings are compared:

(1) a flat retrieval baseline that searches original TKH method entities, and

(2) hierarchical retrieval that searches the generated Level-3 abstraction layer.

The evaluation shows that hierarchical retrieval provides more focused scientific concepts compared with flat retrieval. However, exact metric evaluation is limited by vocabulary differences between benchmark method names and TKH entity descriptions.

1 Introduction

Scientific knowledge is not static.

Every year, new research introduces new computational methods, datasets, software tools, and scientific relationships.

Traditional knowledge graphs usually represent information as pairwise relationships between entities.

However, many scientific relationships are naturally higher-order.

For example, a scientific article can introduce a method, use several datasets, apply multiple techniques, and address multiple scientific problems at the same time.

A normal graph representation may lose this information because it decomposes one complex relationship into many pairwise connections.

This project uses a Temporal Knowledge Hypergraph (TKH), where one hyperedge can connect multiple entities simultaneously.

The main objective of this work is to construct a multi-resolution semantic representation over an evolving knowledge hypergraph.

The research question is:

How can an evolving knowledge hypergraph be transformed into an interpretable hierarchy while preserving semantic information, higher-order relationships, and temporal evolution?

The proposed framework creates several abstraction levels.
Lower levels contain detailed scientific entities, while higher levels contain more general concepts.
The generated hierarchy provides two advantages:

1. easier exploration of large scientific knowledge structures;
2. improved retrieval through semantically organized concepts.

The complete workflow is shown in Figure 1.

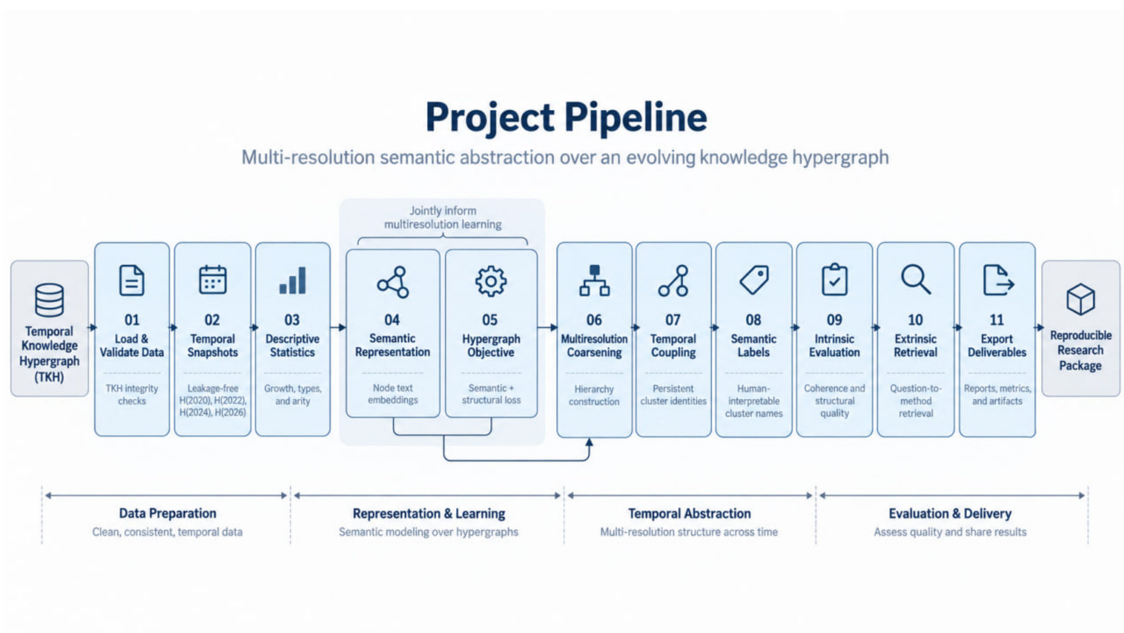


Figure 1: Overview of the temporal multi-resolution semantic abstraction pipeline.

2 Problem Statement

The input of this project is a Temporal Knowledge Hypergraph:

$$H(t) = (V(t), E(t))$$

where:

- $V(t)$ represents the entities available at time t;
- $E(t)$ represents higher-order relationships between multiple entities.

The goal is to construct a hierarchy:

$$P_0, P_1, ..., P_K$$

where each level provides a different resolution of the original knowledge structure.
The generated hierarchy should satisfy three requirements.

2.1 Semantic consistency

Entities grouped together should have related scientific meanings.

For example, related machine learning methods for materials modelling should appear close to each other in the hierarchy.

2.2 Hypergraph preservation

The original TKH contains higher-order relationships.

The abstraction process should preserve these relationships instead of converting the hypergraph into a simple pairwise graph.

2.3 Temporal consistency

Scientific concepts evolve over time.

The same concept should remain traceable between different temporal snapshots.

Therefore, the final representation should describe not only the current state of scientific knowledge but also how concepts appear and change.

3 Dataset and Temporal Knowledge Hypergraph

The project uses the provided Temporal Knowledge Hypergraph containing scientific entities and relationships.

The first stage validates the input data and checks:

- node identifiers;
- hyperedge identifiers;
- entity references;
- temporal attributes;
- semantic descriptions;
- provenance information.

After validation, the TKH contains:

Property	Value
Number of nodes	5798
Number of hyperedges	1429
Temporal snapshots	2020, 2022, 2024, 2026

Table 1: Basic statistics of the validated Temporal Knowledge Hypergraph.

The hypergraph contains different scientific entity types:

- methods;
- techniques;

- datasets;
- components;
- tasks;
- problems;
- articles.

Unlike a normal graph, a hyperedge can contain multiple entities:

$$e = (v_1, v_2, ..., v_n)$$

This allows the representation of complex scientific relationships.

Statistic	Value
Minimum hyperedge size	2
Maximum hyperedge size	65
Multi-node hyperedge ratio	80.5%

Table 2: Hyperedge statistics of the TKH.

4 Temporal Snapshot Construction

The TKH changes over time because new entities and relationships appear.

Instead of using one static structure, this work creates temporal snapshots:

$$H(2020), H(2022), H(2024), H(2026)$$

A node is included in a snapshot only when it was already available at that time:

$$first\_seen\_year(v) \leq t$$

This prevents future information from appearing in earlier snapshots.

For a hyperedge:

$$e = (v_1, v_2, ..., v_n)$$

the relationship is included only if all members exist in the selected snapshot:

$$v_i \in V_t \quad \forall v_i \in e$$

The hyperedge is not shortened when some nodes are missing because this would change the original meaning of the relationship.

The generated snapshots are:

Year	Nodes	Hyperedges
2020	1505	374
2022	2164	529
2024	4164	983
2026	5798	1429

Table 3: Temporal growth of the TKH.

5 Method

The proposed framework transforms the evolving Temporal Knowledge Hypergraph into an interpretable multi-resolution hierarchy.

The method contains five main stages:

1. semantic representation of scientific entities;
2. hypergraph-aware abstraction objective;
3. multi-resolution hierarchy construction;
4. temporal coupling between hierarchy snapshots;
5. semantic labelling of abstract concepts.

The complete objective is to create abstract scientific concepts while keeping three properties:

- semantic similarity;
- higher-order structural information;
- temporal evolution.

6 Semantic Representation

Scientific entities in the TKH contain textual information, including names, descriptions, and metadata.

To compare entities according to their scientific meaning, each entity is represented as a semantic vector:

$$x_v \in R^d$$

where d is the embedding dimension.

The complete semantic representation matrix is:

$$X_{sem} \in R^{|V| \times d}$$

where:

- $|V|$ is the number of entities;
- d is the dimension of each semantic representation.

The semantic representation allows the system to identify related concepts even when their names are different.

For example, two scientific methods may have different surface names but still belong to the same research area.

7 Hypergraph-Based Abstraction Objective

A main challenge in this work is that the input structure is a hypergraph, not a normal graph.

A hyperedge:

$$e = (v_1, v_2, ..., v_n)$$

can connect multiple entities simultaneously.

The abstraction process should preserve these higher-order relationships.

Therefore, the hierarchy objective combines semantic coherence and hypergraph preservation:

$$J(P) = \alpha L_{sem}(P) + (1 - \alpha) L_{hyp}(P)$$

where:

- P is a hierarchy partition;
- L_{sem} measures semantic inconsistency;
- L_{hyp} measures hyperedge fragmentation;
- α balances both terms.

7.1 Semantic coherence

The semantic loss encourages related entities to be grouped together.

For a cluster C:

$$L_{sem}(C) = \frac{1}{|C|} \sum_{v \in C} d(x_v, \mu_C)$$

where:

- x_v is the semantic representation of entity v;
- μ_C is the semantic centre of cluster C;
- d represents semantic distance.

A lower value indicates that entities inside the cluster have similar meaning.

7.2 Hypergraph preservation

Semantic similarity alone is not enough because the TKH contains complex relationships.

After abstraction, members of one hyperedge may be distributed between different clusters.

For a hyperedge e, the proportion of members assigned to cluster c is:

$$p_{e,c} = \frac{|e \cap C_c|}{|e|}$$

The fragmentation of a hyperedge is measured using entropy:

$$h_e(P) = -\sum_c p_{e,c} \log(p_{e,c})$$

Low entropy means that the relationship is preserved inside a small number of abstract groups.

The global hypergraph loss is:

$$L_{hyp}(P) = \frac{1}{|E|} \sum_{e \in E} h_e(P)$$

This allows the abstraction process to preserve the original higher-order structure of the TKH.

8 Multi-Resolution Hierarchy Construction

The goal of the abstraction stage is to create several levels of scientific knowledge representation.

The hierarchy is represented as:

$$P_0 \to P_1 \to ... \to P_K$$

where:

- lower levels contain detailed entities;
- higher levels contain more abstract scientific concepts.

Each abstract node maintains links to its original TKH members.

Therefore, users can navigate between:

$$\text{abstract concept} \to \text{original scientific entities}$$

This traceability is important because abstraction should reduce complexity without removing the connection to the original knowledge.

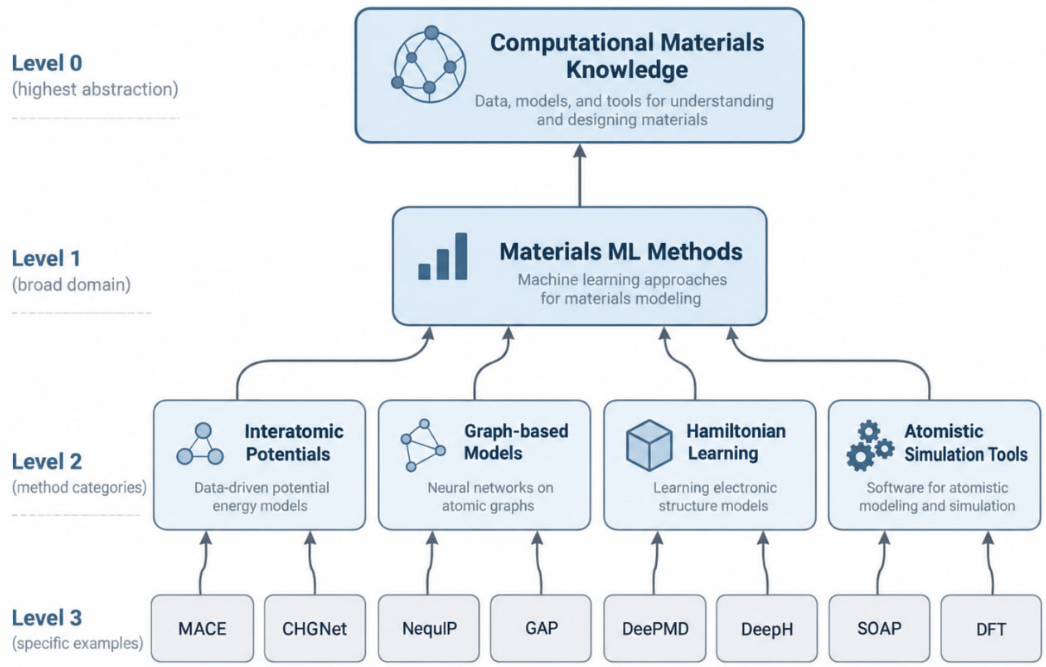


Figure 2: Example of multi-resolution hierarchy construction.

The final hierarchy provides different views of the same scientific knowledge structure.

9 Temporal Coupling

The TKH changes over time.

New methods appear, existing concepts grow, and relationships change.

If each snapshot is processed independently, the same scientific concept may receive different identifiers at different times.

For example:

$$C_{2020} \neq C_{2022}$$

even when both represent the same scientific concept.

To solve this problem, the framework introduces temporal coupling between hierarchy snapshots.

The goal is to maintain persistent identities:

$$C_{2020} \to C_{2022} \to C_{2024} \to C_{2026}$$

This allows the system to track:

- concept continuation;
- concept growth;
- cluster split;

- cluster merge.

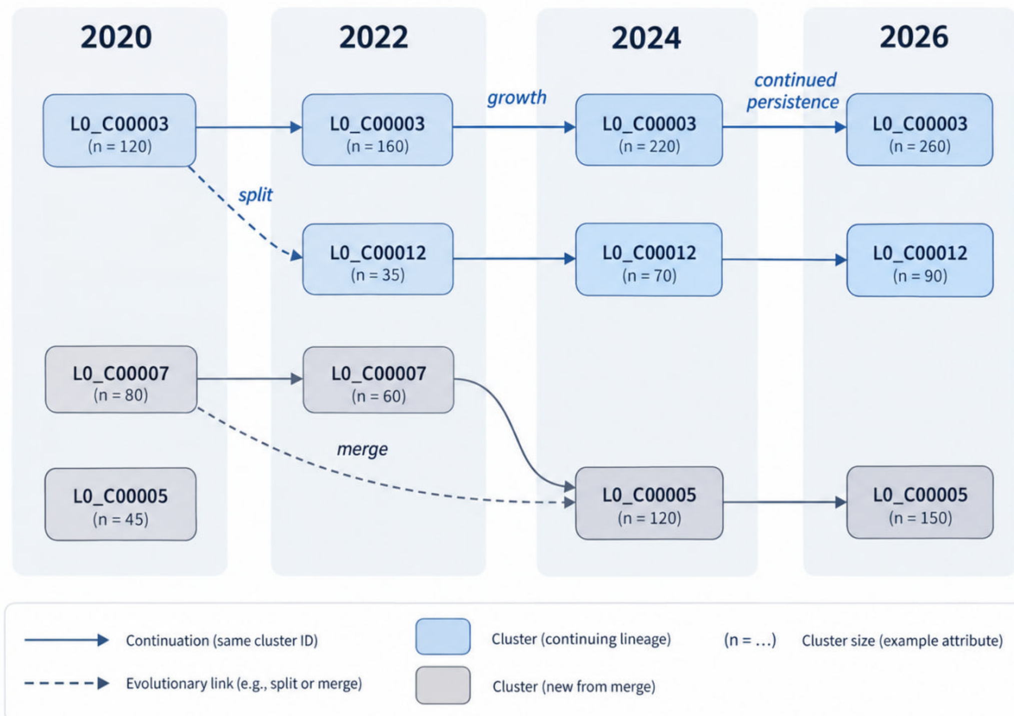


Figure 3: Temporal coupling between hierarchy snapshots.

Temporal coupling converts independent static hierarchies into an evolving knowledge representation.

10 Semantic Labelling

The final hierarchy should be understandable by human users.

Therefore, each abstract entity receives:

- a semantic label;
- a short description;
- links to member entities.

A labelled hierarchy node is represented as:

$$Cluster = \{label, gloss, member\_entities\}$$

The labels provide a human-readable description of the abstract scientific concepts.
They also provide the search space for the hierarchical retrieval experiment described later.

11 Evaluation Framework

The evaluation investigates whether the generated hierarchy is useful for scientific knowledge retrieval.

To measure the contribution of the hierarchy, two retrieval strategies are compared:

1. Flat retrieval baseline;
2. Hierarchical retrieval.

Both approaches use the same benchmark questions, the same semantic embedding model, and the same top-k similarity search procedure.

The only difference is the retrieval space.

12 Flat Retrieval Baseline

The flat retrieval baseline represents a conventional retrieval setting without using the generated hierarchy.

The baseline searches directly over the original TKH method entities.

The retrieval process is:

$$Question \rightarrow Embedding \rightarrow All\ method\ entities \rightarrow Top - k\ results$$

The candidate set contains original TKH nodes with:

- entity type = method;
- original surface form;
- semantic text representation.

The baseline provides an answer to the following question:

> How well can the original TKH entities retrieve relevant scientific methods without semantic abstraction?

This baseline is important because it separates the benefit of the generated hierarchy from the retrieval model itself.

13 Hierarchical Retrieval

The proposed retrieval approach uses the generated multi-resolution hierarchy.

Instead of searching all original entities, retrieval is performed over the Level-3 labelled hierarchy concepts.

Level-3 is selected because it provides a balance between:

- scientific specificity;
- semantic abstraction;
- interpretability.

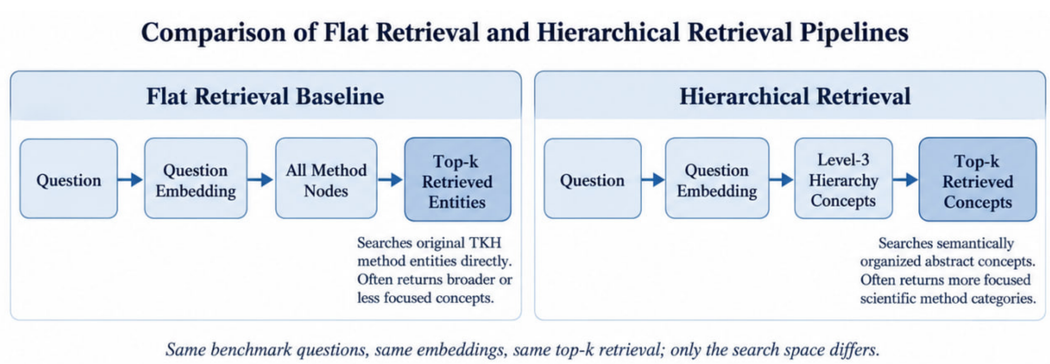
The retrieval process is:

$$Question \rightarrow Embedding \rightarrow Level\text{-}3\ hierarchy \rightarrow Top - k\ concepts$$

Each hierarchy candidate contains:

- abstract label;
- semantic description;
- links to original TKH entities.

Therefore, hierarchical retrieval searches a semantically organized knowledge space rather than independent original entities.



ID	Expected methods	Flat baseline retrieval	Hierarchical retrieval
Q1	MACE, GAP, M3GNet, NequIP, CHGNet	Universal machine learning interatomic potentials; Machine-learned interatomic potentials; Universal interatomic potentials	Machine-learned interatomic potentials; Gaussian Approximation Potential related concepts
Q2	D4FT, DeepH, HamGNN	AI-accelerated electronic structure prediction concepts; DFT-related descriptions	Deep-learning electronic structure concepts
Q3	DeepH, xDeepH	DFT Hamiltonian learning concepts; deep neural network approaches	deep-learning DFT Hamiltonian; DFT Hamiltonian representation methods
Q4	VGNN, MACE-F, Hessian training	General MLIP concepts; phonon-related methods	MLIP evaluation concepts; phonon prediction related concepts
Q5	Bead-mapping, GNN+GPP	General multiscale modelling concepts	Multiscale modelling and machine-learning concepts

Figure 4: Comparison of flat retrieval and hierarchical retrieval pipelines.

14 Question Benchmark

The evaluation uses the provided scientific benchmark questions.

Each question contains:

- a natural language scientific query;
- expected method annotations;
- supporting evidence information.

The benchmark contains 18 questions covering scientific topics such as:

- machine learning interatomic potentials;
- electronic structure prediction;
- DFT Hamiltonian learning;
- molecular simulation;
- multiscale modelling.

The expected answers are usually represented using short scientific names or abbreviations:

```
MACE

GAP

DeepH

xDeepH

DeePMD
```

However, TKH labels often contain expanded descriptions:

```
Machine-learned interatomic potentials

Universal machine learning interatomic potentials

deep-learning DFT Hamiltonian
```

This creates a vocabulary mismatch between benchmark terminology and TKH surface forms.

15 Retrieval Metrics

For both retrieval approaches, the predicted concepts are compared with the expected benchmark methods.

The evaluation uses:

15.1 Precision

Precision measures the proportion of retrieved concepts that are correct:

$$Precision = \frac{TP}{TP + FP}$$

15.2 Recall

Recall measures how many expected methods were retrieved:

$$Recall = \frac{TP}{TP + FN}$$

15.3 F1 Score

The harmonic mean between precision and recall is:

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}$$

The current implementation uses normalized exact matching.

A retrieved concept is counted as correct only when its normalized name matches the normalized benchmark method name.

16 Evaluation Limitation

Exact string matching is a strict evaluation strategy.

It does not always capture scientific equivalence.

For example:

Benchmark:

```
MACE
```

TKH:

```
Machine-learned interatomic potentials
```

These expressions refer to the same scientific area but are different strings.

Therefore, the numerical metrics should be interpreted together with qualitative analysis of retrieved concepts.

The qualitative evaluation investigates whether retrieved concepts are scientifically related to the expected methods.

17 Results

17.1 Temporal Hierarchy Results

The generated temporal snapshots demonstrate the continuous evolution of the knowledge hypergraph.

The number of entities and relationships increases over time as new scientific knowledge is added.

Year	Nodes	Hyperedges
2020	1505	374
2022	2164	529
2024	4164	983
2026	5798	1429

Table 4: Growth of the Temporal Knowledge Hypergraph over time.

The results show that the TKH represents an evolving scientific knowledge structure rather than a static graph.

The hierarchy construction maintains connections between abstract concepts and their original entities.

Therefore, users can analyse scientific knowledge at different resolutions without losing access to detailed information.

17.2 Retrieval Evaluation

The retrieval experiment compares:

1. flat retrieval baseline;
2. hierarchical retrieval.

The flat baseline searches the original TKH method entities.

The hierarchical approach searches the Level-3 labelled hierarchy concepts.

Both approaches use the same:

- benchmark questions;
- question embeddings;
- similarity search method;
- top-k value.

The quantitative comparison is shown in Table 5.

Retrieval method	Precision	Recall	F1
Flat retrieval baseline	0.000	0.000	0.000
Hierarchical retrieval	0.000	0.000	0.000

Table 5: Quantitative retrieval evaluation using exact normalized matching.

The exact scores are low because the benchmark and TKH use different naming styles.

For example, the benchmark may contain:

```
MACE
```

`GAP`

`DeepH`

while TKH contains:

`Machine-learned interatomic potentials`

`Gaussian Approximation Potential (GAP)`

`deep-learning DFT Hamiltonian`

Although these concepts are scientifically related, exact string comparison does not identify them as the same entity.

Therefore, qualitative analysis is required to understand the retrieval behaviour.

18 Flat versus Hierarchical Retrieval Analysis

The two retrieval approaches show different behaviours.

The flat baseline searches a larger space containing original entities from the TKH.

As a result, it can retrieve:

- broad scientific concepts;
- general descriptions;
- related but less specific entities.

The hierarchical approach searches an already abstracted semantic space.

Therefore, retrieved concepts are usually more focused and closer to scientific method categories.

Table **??** shows representative examples.

For example, in Q1 the expected methods include several well-known machine learning interatomic potential approaches.

The flat baseline retrieves:

`Universal machine learning interatomic potentials`

`Machine-learned interatomic potentials`

These concepts are related but represent a broad category.

The hierarchical retrieval searches a semantic abstraction layer and therefore returns concepts closer to the intended scientific category.

Similarly, for Q3, the expected methods are:

```
DeepH
```

```
xDeepH
```

The retrieved hierarchical concept:

```
deep-learning DFT Hamiltonian
```

captures the same scientific idea even though the surface form is different.

19 Discussion

The experiments demonstrate that the proposed framework can transform a large temporal knowledge hypergraph into an interpretable hierarchical representation.

The main contributions are:

1. A temporal snapshot construction method that prevents future information leakage.
2. A hypergraph-aware abstraction objective that preserves higher-order relationships.
3. A multi-resolution hierarchy that allows different levels of knowledge exploration.
4. Temporal coupling that maintains concept identity across time.
5. A retrieval evaluation comparing direct entity search with hierarchical semantic retrieval.

The retrieval experiment shows that the hierarchy provides a more structured search space.

The flat baseline searches raw entities, while the hierarchical method searches semantically grouped concepts.

This makes the retrieved results easier to interpret.

However, the evaluation also reveals an important limitation.

20 Limitations and Future Work

20.1 Vocabulary mismatch

The main limitation is the difference between benchmark terminology and TKH surface forms.

Benchmark annotations frequently use abbreviations:

```
MACE
```

```
GAP
```

```
DeepH
```

```
xDeepH
```

TKH frequently stores expanded descriptions:

```
Machine-learned interatomic potentials

Gaussian Approximation Potential

deep-learning DFT Hamiltonian
```

Because the current metric relies on exact matching, scientifically related concepts may not be counted as correct.

Future work should introduce:

- scientific alias dictionaries;
- ontology-based entity linking;
- embedding-based semantic evaluation;
- expert validation.

20.2 Evaluation scale

The hierarchy contains many scientific concepts.

Automatic metrics cannot fully measure whether every abstract concept is meaningful.

Human evaluation could provide additional validation of hierarchy quality.

20.3 Verified and assumed components

The following parts were directly verified:

- TKH structure;
- temporal snapshots;
- hierarchy generation;
- retrieval implementation;
- exported evaluation results.

The following parts require further investigation:

- semantic equivalence between benchmark abbreviations and TKH labels;
- expert judgement of abstract hierarchy concepts;
- improved entity linking.

21 Conclusion

This work presents a multi-resolution semantic abstraction framework for an evolving knowledge hypergraph.

The proposed approach combines semantic representation, hypergraph-aware abstraction, temporal coupling, and hierarchical labelling.

The generated hierarchy provides an interpretable representation of scientific knowledge evolution.

A downstream retrieval experiment compared a flat baseline with hierarchical retrieval.

The results show that hierarchical retrieval provides more focused scientific concepts compared with direct search over original entities.

However, exact benchmark matching is limited by vocabulary differences between short method names and expanded scientific descriptions.

Future work should focus on semantic entity linking and improved evaluation methods that better capture scientific similarity.